spins but does not expand or contract, and in which one can travel backward in time by simply going out to great distances from Earth and then returning. Physicists object, of course, that our real Universe does not at all resemble Godel's solution: It is *not* spinning, at least not much, and it *is* expanding. In 1976 Frank Tipler used the Einstein field equation to prove that, in order to create a time machine in a finite-sized region of space, one must use exotic material as part of the machine. (Since any traversable wormhole must be threaded by exotic material, the wormhole-based time machines described in this chapter satisfy Tipler's requirement.)

9. In reality, if Carolee were to accelerate up to the speed of light and then back down so quickly, the acceleration would be so great that it would kill her and mutilate her body. However, in the spirit of a physicist's thought experiment, I shall pretend that her body is made of such strong stuff that she can survive the acceleration comfortably.

10. In most science fiction literature, the term "grandfather paradox" is used rather than "matricide paradox." Presumably, the chivalrous men who dominate the science fiction writing profession feel more comfortable pushing the murder back a generation and onto a male.

11. I and my four siblings are very respectful and obedient toward our mother; see, for example, Footnote 2 in Chapter 7. Accordingly, I have sought and received permission from my mother to use this example.

12. Three years later, John Friedman and Mike Morris together managed to prove rigorously that, when waves travel backward in time through a wormhole, there indeed are no unresolvable paradoxes—provided the waves superimpose linearly on themselves in the manner of Box 10.3.

13. See Footnote 8 on page 499.

14. Recently Richard Gott of Princeton University has discovered that one can make a time machine by taking two infinitely long cosmic strings (hypothetical objects that might or might not exist in the real Universe) and moving them past each other at very high speed.